Huawei Access Node WFD-ACC-09 QoS Congestion Event Troubleshooting SOP

Node: WFD-ACC-09 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_8069

Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution steps for a QoS congestion event on Huawei Access Node WFD-ACC-09 at the Whitefield POP. The failure domain of interest is due to Microbursts, interface tail drops, and QoS queue starvation. The Huawei Access Node WFD-ACC-09 is a key component in the access network, responsible for aggregating multiple customer premises equipment (CPE) connections and forwarding traffic towards the core network. The node operates at Layer 2, utilizing the MAC-in-MAC encapsulation technique for QinQ (Double Tagging). The QoS congestion event on this node can be attributed to the misconfiguration of QoS policies, inadequate policing mechanisms, or insufficient buffering capacity. This can lead to packet loss and interface congestion, resulting in the ALARM_8069 alarm being raised. To troubleshoot this issue, it is essential to understand the underlying protocols and hardware mechanics. The Huawei Access Node uses the IEEE 802.1p header for QoS classification, and the QoS policy maps are configured using the ACL (Access Control List) mechanism. Additionally, the node utilizes the policer mechanism to enforce traffic policing. The policer is used to limit the rate of packets transmitted by a particular interface, preventing excessive traffic from overwhelming the node's buffering capacity. In the context of this SOP, the failure domain is a QoS congestion event, which can be caused by a combination of QoS policy misconfiguration, inadequate policing, and insufficient buffering capacity. This document provides a step-by-step guide for troubleshooting and resolving QoS congestion events on Huawei Access Node WFD-ACC-09.

Pre-requisite Checks

- pre-requisite: Verify that the Whitefield POP is accessible and remote hands are available for any necessary physical checks or actions.

- pre-requisite: Check the status of the redundant link to ensure it is active and functioning correctly.

Potential Root Causes

Cause	Probability
QoS policy misconfiguration	High
Inadequate policing mechanisms	Medium
Insufficient buffering capacity	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos policy-map, Expected Output: Policy-map name: , Description: This command is used to display the QoS policy map configuration. It will help identify if there are any misconfigured
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```
QoS policies that may be contributing to the congestion event.

Command: display qos acl, Expected Output: ACL name: , Description:
This command is used to display the QoS ACL configuration. It will
help identify if there are any ACLs that are not correctly mapped to
the QoS policy maps.

Command: display interface congestion, Expected Output: Congestion
status: , Description: This command is used to display the congestion
status of the interface. It will help identify if the interface is
experiencing congestion and if so, what the cause of the congestion
is.
```

Resolution Steps

Step 1: Step: Verify QoS policy map configuration, Command: display qos policy-map, Description: Verify the QoS policy map configuration to ensure it is correctly set up to handle the traffic flows. If the policy map is misconfigured, correct the configuration and re-apply it to the node.

Step 2: Step: Verify QoS ACL configuration, Command: display qos acl, Description: Verify the QoS ACL configuration to ensure it is correctly mapped to the QoS policy maps. If the ACL is not correctly mapped, correct the configuration and re-apply it to the node.

Step 3: Step: Verify policing mechanism configuration, Command: display qos policer, Description: Verify the policing mechanism configuration to ensure it is correctly set up to handle the traffic flows. If the policing mechanism is inadequate, correct the configuration and re-apply it to the node.

Step 4: Step: Verify buffering capacity, Command: display interface buffering, Description: Verify the buffering capacity of the interface to ensure it is sufficient to handle the traffic flows. If the buffering capacity is insufficient, increase the buffering capacity and re-apply it to the node.

Post-Check Verification

```
Command: display qos policy-map, Expected Output: Policy-map name: ,
Description: Verify that the QoS policy map configuration is correct
and correctly applied to the node.

Command: display qos acl, Expected Output: ACL name: , Description:
Verify that the QoS ACL configuration is correct and correctly mapped
to the QoS policy maps.

Command: display interface congestion, Expected Output: Congestion
status: , Description: Verify that the congestion status of the
interface is resolved.
```

Roll-back Procedures

- Step: Revert QoS policy map configuration, Command: undo qos policy-map , Description: Revert the QoS policy map configuration to its previous state if the fix fails.

- Step: Revert QoS ACL configuration, Command: undo qos acl , Description: Revert the QoS ACL configuration to its previous state if the fix fails.

- Step: Revert policing mechanism configuration, Command: undo qos policer , Description: Revert the policing mechanism configuration to its previous state if the fix fails.

- Step: Revert buffering capacity, Command: undo interface buffering, Description: Revert the buffering capacity to its previous state if the fix fails.